

Premkumar M. Patel

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EDUCATION

Sankalchand Patel University

June 2023 – June 2027

Bachelor of Technology in Computer Engineering, CGPA: 7.80 / 10.0

Visnagar, Gujarat, India

- **Coursework:** Data Structures and Algorithms, DBMS, Operating Systems, OOP, Computer Networks, Software Engineering, Machine Learning, Deep Learning

TECHNICAL SKILLS

Languages:	Python, C++, SQL, JavaScript, TypeScript
Backend & APIs:	FastAPI, Flask, RESTful APIs, Pydantic, Webhooks, HMAC Verification
AI & Agent Systems:	LangGraph, LangChain, Autonomous Agents, Tool Calling, LLaMA 3.3 70B, OpenAI API, Groq
Databases & Retrieval:	PostgreSQL, Redis, SQLite, RAG Pipelines, FAISS, Qdrant, Pinecone
DevOps & Tools:	Docker, Git, GitHub, Linux/Bash, Vercel, Render, Postman, UptimeRobot

FEATURED PROJECTS

AgentPay | Python, FastAPI, LangGraph, PostgreSQL, Redis, Next.js, Razorpay API

Feb 2026

- Built an autonomous buyer-agent system where the LLM proposes purchases, but a separate **deterministic policy engine checks and gates every payment** before it reaches the Razorpay API, so the agent can never execute a payment on its own
- Added an append-only **audit trail backed by database triggers** to log every state change, kept merchant data isolated per tenant, and enforced hard spend limits per consumer
- Built a sandbox-to-production pipeline with webhook-driven agent integration, **HMAC signature verification**, and automatic retry with backoff

CampusCircle | Next.js, FastAPI, PostgreSQL, LLaMA 3.3 70B, TypeScript

Nov 2025 – Jan 2026

- Designed and deployed the full backend and AI infrastructure for a verified student platform, scaling it solo to **over 200 active campus users** with average API response times under 150ms
- Shipped three production AI features powered by **LLaMA 3.3 70B**: an automated support agent (“Reva”), an adaptive quiz engine that adjusts question difficulty in real time, and a YouTube-to-study-notes summarizer
- Placed in the **top 18 of over 500 participants** nationwide in The Maverick Effect AI Challenge, run by the Dewang Mehta Foundation Trust

CliniqueAI | Python, Flask, PostgreSQL, Scikit-learn, XGBoost, LangChain, Pydantic

Jan 2026 – Feb 2026

- Trained and benchmarked Logistic Regression, Random Forest, and **XGBoost classifiers** for clinical diabetes risk prediction, tuning the decision boundary for medical explainability and high recall
- Built a **RAG-based patient copilot** grounded on localized EHR records, with transactional CRUD endpoints, Pydantic data schemas, and physician notification alerts
- Removed cold-start latency on Render’s free tier by setting up an automated heartbeat with UptimeRobot

HONORS

National AI Hackathon (Unstop) — Top 20 Finalist, out of 1,600+ teams

Feb 2026

Selected among the top 20 software teams nationwide for CliniqueAI, a diagnostic copilot

National Level

The Maverick Effect AI Challenge — Top 18 Finalist, out of 500+ participants

2026

Recognized by the Dewang Mehta Foundation Trust for CampusCircle, an AI community platform

National Level

CERTIFICATIONS

HackerRank Software Engineer Certification

2025

Problem Solving, SQL, REST APIs